Scalability & Future Plan

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Single crop prediction only, no alternatives shown	Reduces decision flexibility for farmers	Medium
2	Flask development server, not production-hardened	Not suitable for high-traffic public deployment	High
3	No persistent storage/history of past predictions	Cannot track usage trends over time	Low
4	Model trained on a single static dataset	May not generalize to all regional soil types	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance local Flask server	Deploy behind Gunicorn + Nginx, or a managed cloud host (Render/Heroku/AWS)
2	Data Storage	Static CSV dataset	Move to a proper database (PostgreSQL) with support for regional dataset updates
3	Performance	Synchronous single-threaded inference	Add caching for repeated inputs; consider async request handling
4	Security	No authentication (by design)	Add rate-limiting if opened to public API access

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Top-3 crop recommendations with confidence scores	+1 sprint	Gives farmers more informed choices
Phase 3	Fertilizer & irrigation optimization suggestions	+2 sprints	Extends value beyond crop selection alone
Phase 4	Mobile-friendly / regional language support & cloud deployment	+3 sprints	Wider accessibility for real farmers